MILITARY SPECIFICATION

CARTRIDGE, PARACHUTE DEPLOYMENT CCU-11/B

This specification is mandatory for
use by USAF

1. SCOPE

1.1 This specification covers the requirements for one type of cartridge designated as Cartridge, Parachute Deployment, CCU-11/B hereafter referred to as the cartridge. The cartridge is designed to provide a source of gas pressure within the BDU-38/B practice bomb for separation and deployment of the recovery parachute.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-706	Iron Oxide, Ferric, Red Dry Natural and Synthetic
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-A-2550	Ammunition and Special Weapons, General Specification For
MIL-T-7743	Testing, Store Suspension Equipment, General Specification For
MIL-T-13405	Titanium Powder
MIL-D-81303	Design and Evaluation of Cartridges for Stores Suspension Equipment
MIL-C-45227	Components for Propellant Actuated Devices, General Specification For

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Parts
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1167	Lot Numbering of Ammunition
MIL-STD-1168	Ammunition Data Card

DRAWINGS

Air Force

69E14000	Cartridge, Parachute Deployment, CCU-11/B

OTHER PUBLICATIONS

Code of Federal Regulations

49CFR Parts 71-79	Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Material

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Editions Revised 1965) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite 49 CFR Parts 71-79 (Revised 1965).)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Design and Construction. The cartridge shall be constructed in conformance to Air Force drawing 69E14000 and Subsidiary drawings.

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3.2.1 Ignition Composition

3.2.1.1 Ingredients. The ingredients will consist of 53 percent particle size powdered titanium per MIL-T-13405 and 47 percent of reagent grade ACS iron oxide per MIL-I-706.

3.2.1.2 Blending procedure for the above ingredients shall be controlled to satisfy all electrical and performance requirements when assembled in the CCU-11/B cartridge.

3.3 Function. The function of the cartridge is to provide a source of gas pressure for separation and deployment of the recovery parachute from the BDU-38/B practice bomb.

3.3.1 Ballistics. The cartridge installed in the inner deployment tube of the BDU-38/B practice bomb shall produce performance as noted below:

	Velocity	Stroke
-65 ± 5°F	*	50.5 in.
+70° ± 5°F	155 ± 20 ft/sec	50.5 in.
+325 ± 5°F	*	50.5 in.

*Performance values at these temperatures are to be recorded for information only.

3.3.2 Before motion of the parachute deployment system occurs, four (4) shear plates holding the tail cap to the aft section of practice weapon shall be sheared. Force to overcome this load shall be 11,000 + 2,000 pounds.

3.3.3 The cartridge shall perform as noted in 3.3.1 and 3.3.2 while accelerating a weight of 105 ± 5 pounds, consisting of the outer deployment tube assembly, parachute, deployment plate, thermal cup, and rear extension of the practice weapon. Total system performance shall not exceed 200 milliseconds maximum.

3.4 Performance

3.4.1 All-Fire Sensitivity. The cartridge shall fire within 50 milliseconds, under all environmental conditions, when subjected to a direct current of 5 amperes at 24-28 volts d.c.

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3.4.2 No Fire Sensitivity. The cartridge shall be capable of withstanding a minimum direct current of 1 ampere or a minimum power of 1 watt for a duration of five minutes, minimum, without firing under any environmental conditions as specified in the No-fire sensitivity test of this specification.

3.5 Electric Characteristics

3.5.1 Ignition Element Resistance. The resistance of each bridge circuit shall be 1.1 ohms, plus or minus 0.2 ohm.

3.6 Environmental Requirements

3.6.1 Static Discharge. The cartridge shall not fire when subjected to an electrostatic discharge of 150,000 ergs from a 500 picofarad capacitor.

3.6.2 Stray Voltage. The cartridge shall be capable of proper performance after subjection to 2000 pulses of direct current. Each pulse shall be 105 $\pm$ 5 milliamperes. Duration of pulse shall be 300 milliseconds and pulse recurrent rate shall be 2.0 pulses per second.

3.6.3 Vibration. The cartridge shall be capable of satisfactory operation during and after vibration, as specified in vibration tests of this specification.

3.6.4 Airtightness. The cartridge shall not exhibit a leak rate of greater than 1×10^{-4} cc/sec when subjected to a dry leak test at a pressure differential of 14 psi.

3.6.5 Temperature. The cartridge shall be capable of proper performance during and after prolonged exposure at temperature from -65°F to +325°F.

3.6.6 Temperature Shock. The cartridge shall be capable of proper performance after being subjected to the temperature shock testing requirements of this specification.

3.6.7 Impact Shock. The cartridge shall be capable of proper operation after being subjected to a 6-foot drop onto a steel plate supported by concrete. The cartridge shall not fire and be safe to handle after being subjected to a 10-foot drop onto a steel plate supported by concrete.

3.6.8 Temperature and Humidity Cycling. The cartridge shall be capable of proper performance after being subjected to the temperature humidity cycling test contained in this specification. At the conclusion of the test, cartridges must meet functional requirements and be free from visible damage.

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3.6.9 Protective Finish. The protective finish shall be in accordance with the applicable drawings and specifications, and shall be capable of withstanding the salt spray exposure as required by this specification.

3.7 Workmanship. The cartridge, including all parts and accessories, shall be fabricated in a manner that will ensure that the criteria of appearances, fit, and adherence to tolerances, as specified on the drawings, shall be observed. Particular attention shall be given to the markings required under "Identification of Product".

3.8 Identification of Product. The cartridge shall be marked for identification in accordance with MIL-STD-130.

3.9 Lot Numbering. Each separate production lot shall be assigned an ammunition lot number in accordance with MIL-STD-1167. This lot number shall be marked on each cartridge, as shown on the assembly drawing.

3.10 Storage Group. The cartridge belongs to storage compatibility Groups B, E, and M.

3.11 Quantity Distance Classification. The cartridge is a Class 3 Quantity-Distance item.

3.12 ICC Explosive Classification. CCU-11/B Cartridges shall be assigned ICC explosive classification in accordance with 49 CFR Parts 71-79.

3.13 Government Loaned Property. The following item shall be loaned to the contractor, upon his request, as Government Loaned Property for the purpose of testing the cartridge.

Item Number	Part Number	Description of Item	Quantity
1	68D14398	Parachute Deployment	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. The inspection and testing of the cartridge shall be as follows:

a. Preproduction Tests

b. Acceptance Tests

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4.2 Preproduction Testing

4.2.1 Preproduction Testing. The preproduction test samples shall be representative of production cartridges. They shall be consecutively numbered for test identification. They shall be tested at a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's plant under the supervision of the procuring activity. All samples shall be tested in a BDU-38/B practice bomb or approved functional substitute test fixture.

4.2.2 Preproduction Test Report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all test described in "4.5 Test Methods" as listed in Table I and Table II.

4.3 Acceptance Tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Plans and Tests

4.3.1 Individual Tests. Each cartridge shall be subjected to the individual tests listed in Table I.

4.3.2 Sampling Plan and Tests

4.3.2.1 Lot. Lot definition, size and formation shall be in accordance with MIL-STD-105 and shall contain:

a. Components produced by a homogenous process.

b. Propellant from not more than one lot.

c. Ignition composition from not more than one lot.

d. Ignition elements from not more than one lot.

4.3.2.2 Sampling Plan. The sample size shall be in accordance with MIL-STD-105, Inspection Level II and Table II-A, AQL of 1.0. Sample tests will be as stated in Table I of this specification. A sampling test plan shall be submitted to the procuring activity for approval before production begins.

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TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	PREPRO-DUCTION TEST	INDIVIDUAL TEST	SAMPLING TEST
Examination of Product	3.2	4.5.2	X	X	
Conformance to Drawings	3.2	4.5.3	X	X	
X-ray Examination		4.5.4	X		X
All Fire Sensitivity	3.4.1	4.5.5	X		X*
No Fire Sensitivity	3.4.2	4.5.6	X		X
Ignition Element Resistance	3.5.1	4.5.7	X	X	
Static Discharge	3.6.1	4.5.8	X		
Stray Voltage	3.6.2	4.5.9	X		
Vibration	3.6.3	4.5.10	X		X
Airtightness	3.6.4	4.5.11	X		X
High Temperature	3.6.5	4.5.12	X		X*
Low Temperature	3.6.5	4.5.13	X		X*
Temperature Shock	3.6.6	4.5.14	X		
Impact Shock	3.6.7	4.5.15	X		
Temperature Humidity	3.6.8	4.5.16	X		
Ballistic Firing	3.3.1		X		X

*To be conducted in conjunction with ballistic firings.

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TABLE II

PREPRODUCTION SCHEDULE AND TEST PLAN

Test Title									
Air Tightness	1-90[1]								
X-ray	1-90[1]								
Electric Resistance	1-90[1]								
Environmental Tests									
Temperature Shock		46-51							85-90
Vibration			52-57						85-90
Temperature-Humidity				58-63					85-90
Impact Shock					64-72[2]				85-90[3]
Static Discharge						73-78			85-90
Stray Voltage							79-84		85-90
Performance Firings									
No-Fire Sensitivity	1-90								
All-Fire Sensitivity	1-90[4]								
Cartridge Firings -65°F	1-15	46-47	52-53	58-59	64-65	73-74	79-80	85-86	
Cartridge Firings +70°F	16-30	48-49	54-55	60-61	66-67	75-76	81-82	87-88	
Cartridge Firings +325°F	31-45	50-51	56-57	62-63	68-69	77-78	83-84	89-90	

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4.3.3 Rejection and Retest. When a sample cartridge fails to meet specified test, no cartridge on hand or later produced shall be accepted until the extent and cause of failure are determined. The cause of failure shall be corrected and subsequent cartridge shall demonstrate compliance by passing the failed tests.

4.3.4 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions

4.4.1 Standard Conditions. Test conditions shall be as specified in each specific test under "4.5 Test Methods." If no conditions are stated, the following shall apply:

a. Temperature 21°C (70°F) ± 11°C (20°F)

b. Relative Humidity - 90 percent or less.

4.4.2 Test Temperature. The cartridge shall be stabilized at the specified temperature prior to conducting any tests. Temperature stabilization shall be as defined in MIL-STD-810.

4.4.3 Test Measurements

4.4.3.1 Instruments. All measurements shall be made with instruments whose accuracy has been verified. Calibration records shall be maintained.

4.4.3.2 Tolerances. The maximum allowable tolerances on test condition measurements shall be as follows:

a. Temperature - plus or minus 2.8°C (5°F).

b. Relative Humidity - plus or minus 5 percent.

c. Vibration and Shock Amplitude - plus or minus 5 percent.

d. Vibration Frequency - plus or minus 2 percent.

e. Current - plus or minus 2 percent.

f. Resistance - plus or minus 2 percent.

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g. Time Interval - plus or minus 2 percent.

h. Pressure - plus or minus 5 percent.

i. Reaction Force - plus or minus 5 percent.

j. Velocity - plus or minus 5 percent.

4.4.4 Required Measurements

a. Resistance of unit prior to firing.

b. Firing current.

c. Firing time.

d. Pressure vs time.

e. Reaction force vs time.

f. Velocity

g. For preproduction testing, the cartridge number shall be recorded.

h. Other measurements shall be taken as specified in preproduction schedule and test plan.

4.5 Test Methods

4.5.1 Classification of Tests. Inspection and testing of the cartridge shall be as shown in Table I.

4.5.2 Examination of Product

4.5.2.1 Examination of Components. One hundred percent examination shall be performed for all drawing characteristics when the component lot size is not in excess of 300 cartridges. For lot sizes in excess of 300 cartridges, sampling inspection will be performed utilizing a plan acceptable to the procuring agency, in accordance with MIL-C-46227. Failure to comply with all drawing and specification requirements shall be cause for rejection.

4.5.2.2 Examination of Assemblies. One hundred percent examination shall be performed to determine drawing compliance with all requirements of the assembly drawing and this specification when the

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assembly lot size is not in excess of 300 cartridges. For lot sizes in excess of 300, sampling inspection will be performed to determine drawing compliance utilizing a plan acceptable to the procuring agency in accordance with MIL-C-46227.

4.5.3 Conformance to Drawings. Each cartridge shall be inspected for strict conformance to the engineering drawings, and to specifications and standards referenced therein. Failure to meet these criteria shall be cause for rejection.

4.5.4 X-Ray Examination. Each cartridge of the preproduction test sample shall be subjected to X-ray examination. X-ray photographs shall be identified by date, cartridge part numbers, lot number and serial numbers. These X-ray photographs shall be examined to determine compliance with the requirements of this specification.

4.5.5 All Fire Sensitivity. Cartridge shall be tested according to Table II. Failure of any cartridge to fire, or failure to meet the ballistic requirements of this specification, shall be cause for rejection.

4.5.6 No Fire Sensitivity. Cartridges shall be tested according to Table II. The cartridges shall be held in a test jig and tested as follows:

Bridge Circuit Resistance	Requirement (Min)
one ohm	one ampere or one watt
one ohm	one watt
one ohm	one ampere

Each bridge wire circuit shall receive this test. Firing of the cartridge during the five minute test shall be cause for rejection.

4.5.7 Ignition Element Resistance. The resistance of each bridge circuit shall be measured within an accuracy of one percent using a test circuit which subjects the bridge circuit to a current of less than 50 milliamperes.

4.5.8 Static Discharge. Twelve (12) cartridges shall be tested to the static discharge test and shall not fire when subjected to an electrostatic discharge of 150,000 ergs from a 500 picofarad capacitor.

4.5.9 Stray Voltage. Each of twelve (12) cartridges shall be subjected to 2000 pulses of direct current. Each pulse shall be 100 milliamperes. Duration of pulse shall be 300 milliseconds and pulse recurrent rate shall be 2.0 pulses per second.

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4.5.10 Vibration. Twelve (12) cartridges shall be subjected to vibration tests prescribed in MIL-STD-810, Method 514, Procedure I, except that Figure I and Table III of this document apply in lieu of Figure I and Table I in MIL-STD-810. Equal quantities shall be vibrated at -65°F, room temperature and +250°F after a one hour soak time. Times shown below refer to one axis of vibration.

TABLE III

VIBRATION TEST SCHEDULE

NUMBER OF RESONANCES	TOTAL TIME AT RESONANCE	TOTAL CYCLE TIME
0	10 minutes	60 minutes
1	10 minutes	50 minutes
2	20 minutes	40 minutes
3	30 minutes	30 minutes
4	40 minutes	20 minutes

4.5.11 Airtightness. Ninety (90) cartridges shall be tested in accordance with Table I and shall not exhibit a leak rate of greater than 1×10^{-4} cc/sec when subjected to a dry leak test at a pressure differential of 14 psi. Tests shall be conducted using a dry gas leak detection of sufficient sensitivity to ascertain if the cartridge meets the leak rate requirements.

4.5.12 High Temperature Operation. Twenty-nine (29) cartridge shall be conditioned at a temperature of 163°C (325°F) for at least four hours. Cartridges shall be fired within three minutes after removed from the temperature chamber.

4.5.13 Low Temperature. Twenty-nine (29) cartridges shall be conditioned to, and maintained at a temperature of -54°C (-65°F) for at least four hours. Cartridges shall be fired within three minutes after removed from the temperature chamber.

4.5.14 Temperature Shock. Twelve (12) cartridges shall first be placed within a test chamber wherein a temperature of 111°C (250°F) is maintained. The cartridges shall be exposed to this temperature for a period of four (4) hours, at the conclusion of which, and within five (5) minutes, the equipment shall be transferred to a chamber having an internal temperature of -64°C (-65°F). The equipment shall be subjected to this temperature for a period of four (4)

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RANGE CURVE FOR VIBRATION TEST

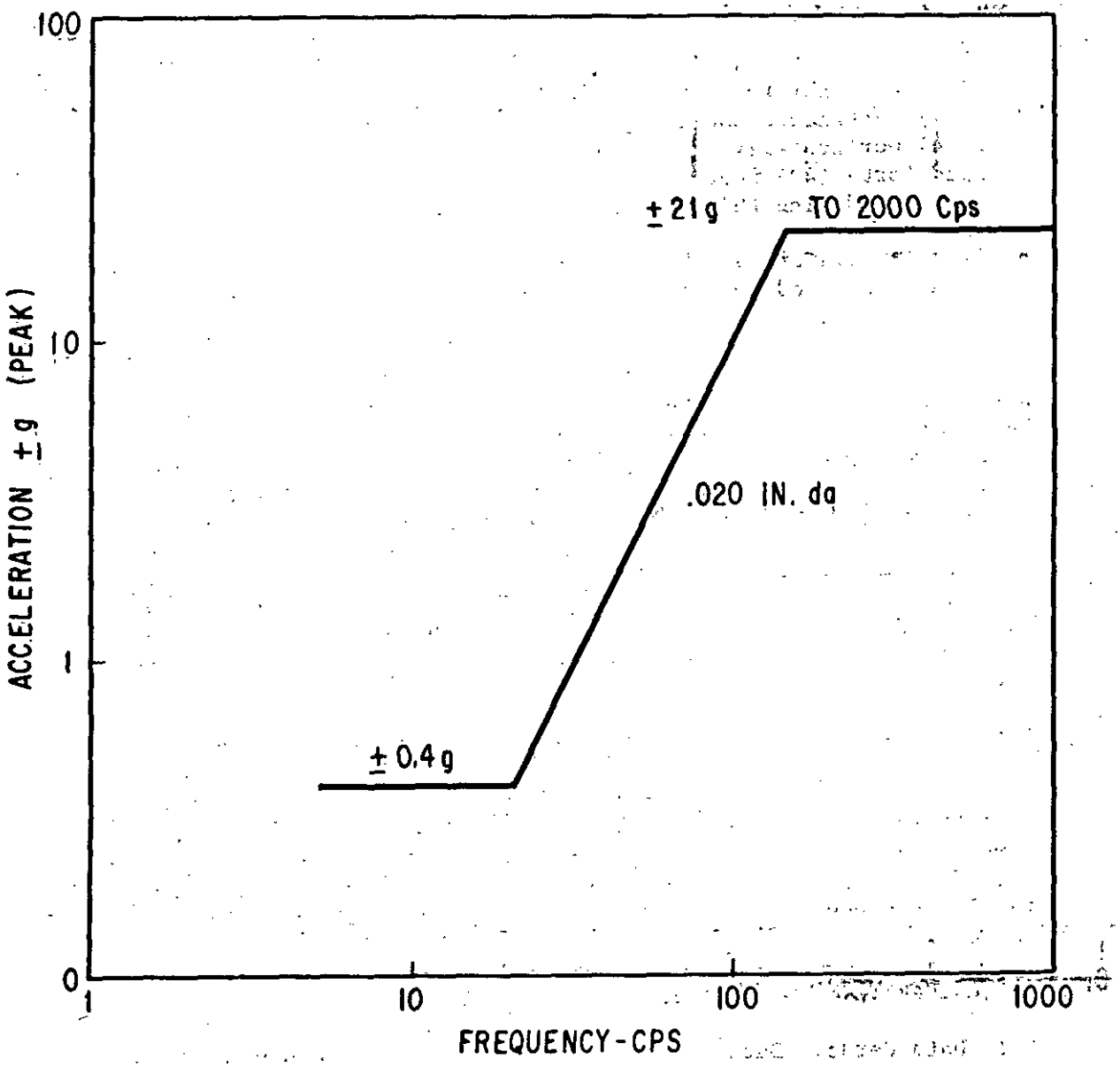


FIGURE 1

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hours. This constitutes one (1) cycle. The number of complete cycles shall be three (3). The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the cartridges shall be removed from the test chamber, conditioned, and tested according to Table I.

4.5.15 Impact Shock. Twelve (12) cartridges shall be dropped from a height of eight feet onto a steel plate supported by concrete. Four (4) cartridges shall impact nose down, four (4) nose up, and four (4) horizontally. In addition three (3) cartridges shall be dropped forty (40) feet and land in any impact position. Firing of a cartridge during this test shall be cause for rejection.

4.5.16 Temperature and Humidity Cycling. Twelve (12) cartridges shall be subjected to this test in accordance with MIL-D-81303 paragraph 4.2.5.

4.5.17 After the environmental tests the 42 cartridges shall be x-rayed, air tightness and checked for electrical resistance and ballistically tested. Evidence of any disintegration of the ignition element, igniter or propellant or derangement of any component or failure to meet the electrical resistance requirements or ballistics performance shall be cause for rejection of the lot.

4.5.18 Temperature Conditioning. Ballistic firings will be conducted in accordance with table II. Cartridges shall be soaked at the designated temperature for at least four hours. Cartridges shall be fired within three minutes after removed from the temperature chamber.

4.6 Test Procedures shall be in accordance with MIL-T-7743.

4.6.1 Instrumentation. The instrumentation used to measure the current flow is to have an accuracy of $\pm$ 1 percent. When a rectified current is used the root mean square ripple content shall not exceed 5 percent of the test current and the rise time of the pulse shall not exceed 50 microseconds with a maximum overshoot of 5 percent.

5. PREPARATION FOR DELIVERY

5.1 Data Cards. Data card information shall be as specified in MIL-STD-1167.

5.2 Packaging. For all levels of shipment, cartridges shall be packaged as specified by the procuring activity.

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5.3 Marking. Exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

CARTRIDGE, PARACHUTE DEPLOYMENT, CCU-11/B

5.3.1 Additional Marking (Explosive). In addition to marking in accordance with MIL-STD-129, the exterior containers shall be marked in accordance with Code of Federal Regulations 49CFR71-79. The following specific warning shall appear on the exterior container: "ELECTRIC EXPLOSIVE CARTRIDGES, Handling With Care".

6. NOTES

6.1 Intended Use. The Cartridge Parachute Deployment, CCU-11/B is intended for use in the BDU-38/B Practice Bomb for separation and deployment of recovery parachute or in similar applications where a source of high pressure gas generation is required.

6.2 Ordering Data. The procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaned property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procurement activity.

6.2.1 Preproduction Samples. It is expected that the contract or purchase order will specify that 90 cartridges, representative of the production cartridges will be submitted to the contractor or designated laboratory for test and evaluation. The invitation for bids, and the contract or purchase order, should specify the point of delivery for these items.

6.3 Production Delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions.

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

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b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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TABLE II (Cont'd)

1 - Tests will be repeated following the environmental test phase.

2 - Use No.'s 70, 71, & 72 for forty foot drop test. Discard after test.

3 - Do not perform forty foot drop test.

4 - To be performed with ballistic firings.

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